Jiuqi Wang

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Education

University of Virginia *Aug. 2023 - Sept. 2027*
Ph.D. Computer Science Charlottesville, Virginia
Advisor: Shangtong Zhang

University of Alberta *Sep. 2021 - Jun. 2023*
MSc. Computing Science (Thesis) Edmonton, Alberta
Co-Supervisors: Martin Müller and Jonathan Schaeffer

McGill University *Sep. 2017 - Jun. 2021*
BSc. Honours Computer Science Montréal, Québec
Distinction, First-Class Honours

Research Interests

Reinforcement Learning, In-Context Learning, Agentic Systems

Publications

* indicates equal contribution; † indicates advisor

Peer-Reviewed

1. Predicting Plasticity in Deep Continual Learning: A Theoretical Perspective
   Jiuqi Wang, Jayanth Srinivasa, Claire Chen, Shuze Daniel Liu, Ali Payani, Shangtong Zhang†
   Poster at **Neural Information Processing Systems (NeurIPS)**, 2026
2. InfRL: Inference-time Reinforcement Learning for Research Idea Optimization
   Sikun Guo, Amir Hassan Shariatmadari, **Jiuqi Wang**, Albert Huang, Stefan Bekiranov, Shangtong Zhang†, Aidong Zhang
   Knowledge Discovery and Data Mining (KDD) AI for Sciences Track, 2026
3. Almost Sure Convergence of Linear Temporal Difference Learning with Arbitrary Features
   Jiuqi Wang, Shangtong Zhang†
   Journal of Machine Learning Research (JMLR), 2026
4. Almost Sure Convergence of Differential Temporal Difference Learning for Average Reward Markov Decision Processes
   Ethan Blaser, **Jiuqi Wang**, Shangtong Zhang†
   Poster at **Artificial Intelligence and Statistics (AISTATS)**, 2026
5. Towards Provable Emergence of In-Context Reinforcement Learning
   Jiuqi Wang, Rohan Chandra, Shangtong Zhang†
   Poster at **Neural Information Processing Systems (NeurIPS)**, 2025
6. Transformers Learn Temporal Difference Methods for In-Context Reinforcement Learning
   Jiuqi Wang*, Ethan Blaser*, Hadi Daneshmand, Shangtong Zhang†
   Poster at **International Conference on Learning Representations (ICLR)**, 2025
   Contributed Talk at the **RLC Workshop on Training Agents with Foundation Models**, 2024
   Spotlight Award at the **ICML Workshop on In-Context Learning**, 2024
7. Deep Dive on Checkers Endgame Data
   Jiuqi Wang, Martin Müller†, Jonathan Schaeffer†
   Oral presentation at **IEEE Conference on Games (CoG)**, 2023

Preprint

1. Rethinking the Design of Reinforcement Learning-Based Deep Research Agents
   Yi Wan*, **Jiuqi Wang***, Liam Li, Jinsong Liu, Ruihao Zhu, Zheqing Zhu
   arXiv:2510.15862, 2026
   Work done during internship at *Pokee AI*; Widely adopted open-source release accompanying the paper.
2. Experience Replay Addresses Loss of Plasticity in Continual Learning
   Jiuqi Wang, Rohan Chandra, Shangtong Zhang†
   arXiv:2503.20018, 2025
3. A Survey of In-Context Reinforcement Learning
   Amir Moeini, **Jiuqi Wang**, Jacob Beck, Ethan Blaser, Shimon Whiteson, Rohan Chandra, Shangtong Zhang†
   arXiv:2502.07978, 2025

Research & Professional Experience

Student Researcher (Upcoming) *May 2026 – Nov. 2026*
Google DeepMind London, UK

- Student Researcher in the autonomous agents team hosted by Martin Klissarov and Hao Sun.

Research Scientist Intern *Jun. 2025 – Dec. 2025*
Pokee AI Remote

- Co-led the PokeeResearch-7B open-source project end-to-end.
- Leveraged verl to efficiently RL-fine-tune and experiment with PokeeResearch-7B.
- Developed tool-server infrastructure and tool-calling schema to support web search and content retrieval.
- Built deployment tools (CLI + Gradio) for model accessibility and evaluation automation.
- Achieved state-of-the-art performance among 7B-class models on a range of modern Q&A benchmarks; 1.1k stars & 400+ forks on GitHub (as of Nov. 2026).

Graduate Research Assistant *Aug. 2023 - present*
Advised by Prof. Shangtong Zhang Charlottesville, Virginia

- Developed theoretical foundations for In-Context Reinforcement Learning (ICRL) in Transformers, resulting in papers accepted at ICLR 2025 and NeurIPS 2025.
- Proved almost-sure convergence of TD learning with arbitrary features; journal paper currently under review.

Master's Thesis – Deep Dive on Checkers Endgame Data *May 2022 - Jun. 2023*
Co-Supervised by Prof. Martin Müller and Jonathan Schaeffer Edmonton, Alberta

- Designed neural network and training pipeline for knowledge compression of checkers endgame data with JAX/Haiku.
- Published as a full paper (oral presentation) at CoG, 2023.

Development Intern *May 2021 – Jul. 2021*
PTC Inc. Suzhou, China

- Improved product internationalization by integrating i18next for multi-language support.

Undergraduate Research *May 2020 - Sep. 2020*
Supervised by Prof. Hsiu-chin Lin Montréal, Québec

- Built and evaluated sim-to-real transfer learning for LPV dynamical systems using Gaussian Mixture Models.
- Developed adaptive retraining procedure to improve robustness in learning with noisy data.

Teaching Assistantships

- *Reinforcement Learning (Spring 2026, Fall 2024, University of Virginia)*: hosted office hours and graded assignments.
- *Search, Knowledge and Simulations (Spring 2022, University of Alberta)*: prepared assignment starter code and auto-grading scripts; organized and evaluated the final project (game-playing agent tournament).
- *Intro to the Foundations of Computation I (Fall 2021, University of Alberta)*: held lab sessions to review and complement the course material.
- *Intro to Computer Science (Spring 2020, McGill University)*: prepared original weekly quiz questions.

Volunteering Experience

High School Mentor *Sep. 2024 - Dec. 2024*
Charlottesville High School Charlottesville, Virginia

- Matched with a team in the senior engineering class and advised their final engineering project.
- Held weekly meetings with the team to keep up with the progress and assisted troubleshooting.

Peer Review

NeurIPS (2025-2026), EWRL (2026), ICML (2024–2026), ICLR (2024–2026), AISTATS (2024-2026), IJCAI (2025-2026), AAMAS (2025–2026), ACML (2024), RLC (2024–2026)

Honors and Awards

Best Reviewer Award *Apr. 2025*
AISTATS

AI-SCORE Summer School Fellow *May 2024*
$900 University of Maryland, College Park

IEEE CIS Travel Grant *Jul. 2023*
$500 IEEE CIS

GSA Academic Travel Grant *Jul. 2023*
$500 University of Alberta

Science Undergraduate Research Award (SURA) *May 2020*
$7,000 McGill University

Dean's Honour List *Aug. 2018*
McGill University

Merit-Based Undergraduate Scholarships (Condensed) *Jul. 2017 – Jul. 2018*
$6,500 McGill University

Skills

- **Technology:** Python (NumPy, PyTorch, JAX, vLLM, Gradio), Java, Git/GitHub, Docker, Linux, Weights & Biases, Huggingface, LaTeX
- **Knowledge:** Statistical Machine Learning, Deep Learning, (Deep) Reinforcement Learning, Heuristic Search, Robotics
- **Language:** English (proficient), Mandarin (native), French (basic)